



18th Iranian Statistics Conference
K.N.Toosie University of Technology

29-31 August 2026



A Robust FSVM-GARCH Framework for Imbalanced Time Series Data

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Abstract:

This paper presents an enhanced fuzzy support vector machine (FSVM) framework tailored for financial time series classification. We address the persistent challenges of noise and class imbalance by introducing a novel integration of domain-based volatility information and adaptive fuzzy weighting. The key innovation lies in incorporating the Parkinson volatility estimator—a highly efficient and model-free variance measure—as the sole input feature, ensuring robustness in environments where high-frequency data are unavailable. Furthermore, we develop a weighted fuzzy membership mechanism (WFSVM) that assigns membership values based on inverse class frequency weighting and volatility-based distance metrics. This dual mechanism significantly enhances the classifier's performance, allowing it to outperform standard SVM and state-of-the-art imbalance-corrected models in volatility regime recognition, especially under skewed data distributions.

Keywords: Fuzzy Support Vector Machine (FSVM), Class Imbalance Learning (CIL), Parkinson Volatility Estimator, Volatility Clustering, Mahalanobis Distance

Mathematics Subject Classification (2010): 62H30, 91G70, 68T10.

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1 Introduction

Classification of financial time series remains a critical task in econometrics and data mining, essential for risk management and investment decision-making. Support Vector Machines (SVM), introduced by Vapnik [Vapnik \(1995\)](#), have become a cornerstone technique due to their ability to handle high-dimensional data and minimize the upper bound of generalization error [Boser et al. \(1992\)](#). Despite their success in cross-sectional analysis, standard SVMs often struggle with the unique characteristics of financial data, specifically noise, non-stationarity, and class imbalance. Financial markets are characterized by volatility clustering and heteroskedasticity, where the distribution of classes (e.g., high volatility vs. low volatility regimes) is often significantly skewed [Andersen and Bollerslev \(1998\)](#).

To mitigate the impact of noise and outliers, Fuzzy Support Vector Machines (FSVM) were proposed by Lin and Wang [Lin and Wang \(2002\)](#). FSVM assigns a fuzzy membership score to each sample, reducing the influence of outliers on the decision boundary. However, standard FSVM does not inherently address class imbalance. In financial contexts, minority classes often represent critical market events, making their accurate detection paramount. Recent approaches, such as Batuwita and Palade [Batuwita and Palade \(2010\)](#) and Wu et al. [Wu et al. \(2017\)](#), have integrated sampling techniques and penalty weights to balance classes. Furthermore, advancements in kernel functions and loss functions, such as the pinball loss and self-tuned schemes, have improved robustness [Verma et al. \(2023\)](#); [Tanveer et al. \(2024\)](#).

Another line of research combines machine learning with econometric models. For instance, Khemchandani et al. [Khemchandani et al. \(2009\)](#) demonstrated the efficacy of SVR in financial forecasting. Zhang and Wang [Zhang and Wang \(2009\)](#) integrated fuzzy logic with GARCH models. While these hybrid models capture temporal dependencies, they often rely on traditional volatility proxies like squared returns, which are notoriously noisy [Degiannakis and Livada \(2013\)](#). This paper proposes a robust framework that bridges these gaps. We integrate GARCH modeling with a Weighted FSVM (WFSVM) but introduce a critical innovation: the use of the Parkinson volatility estimator [Parkinson \(1980\)](#) as the primary feature. By coupling this efficient estimator with a Mahalanobis-distance-based fuzzy membership function, we create a classifier that is robust to noise, adaptive to class imbalance, and efficient in low-frequency data environments.

2 Proposed Framework

Our methodology consists of two primary phases: robust feature extraction using Parkinson volatility and classification via a Weighted Fuzzy SVM that incorporates class imbalance learning (CIL).

2.1 Volatility Estimation: GARCH and Parkinson

Financial time series exhibit volatility clustering, where large changes tend to be followed by large changes.

The GARCH(1,1) model is a standard tool for capturing this phenomenon:

$$h_t = w + \alpha Z_{t-1}^2 + \beta h_{t-1}, \quad Z_t = e_t \sqrt{h_t}, \quad e_t \sim IID(0, 1) \quad (2.1)$$

While h_t captures the conditional variance, estimating it directly requires complex model fitting and assumptions. A more practical and efficient proxy for our classification task is the **Parkinson estimator**, which utilizes the high and low prices of the trading day [Parkinson \(1980\)](#):

$$\sigma_{park,t}^2 = \frac{1}{4 \ln 2} \left[\ln \left(\frac{H_t}{L_t} \right) \right]^2 \quad (2.2)$$

Theoretically, the Parkinson estimator is significantly more efficient (under a Brownian motion assumption) than the squared return proxy $\sigma_t^2 = r_t^2$ [Molnar \(2016\)](#). Unlike realized variance, which requires high-frequency data, the Parkinson estimator relies only on daily OHLC data, making it widely applicable [Mapa \(2003\)](#). In our framework, we construct the feature vector x_i solely from this estimator:

$$x_i = [\sigma_{park,t}^2] \in \mathbb{R}^1 \quad (2.3)$$

This simplification focuses the model on the most informative volatility feature without introducing model-dependent biases.

2.2 Weighted Fuzzy SVM (WFSVM) for Imbalanced Data

Standard SVM seeks an optimal hyperplane $f(x) = w^T x + b$ by solving the dual optimization problem with penalty parameter C . To handle noise, FSVM introduces a membership value m_i that modulates the penalty term for misclassification ζ_i [Lin and Wang \(2004\)](#):

$$\min_{w,b,\zeta} \left[\frac{1}{2} \|w\|^2 + C \sum_{i=1}^t m_i \zeta_i \right], \quad s.t. \quad y_i (w^T x_i + b) \geq 1 - \zeta_i \quad (2.4)$$

We propose a novel definition for m_i that combines class-level balancing with sample-level relevance:

$$m_i = f(x_i) \pi_j \quad (2.5)$$

where π_j is the weight assigned to class j to handle imbalance, and $f(x_i)$ represents the sample's membership within that class.

Class Imbalance Weighting (π_j):

To counter the bias towards majority classes, we sort class sizes $n_{(1)} \leq n_{(2)} \leq \dots \leq n_{(K)}$. We assign

the largest weight to the smallest class and vice-versa, inspired by inverse frequency weighting strategies [Diez-Pastor et al. \(2015\)](#):

$$\pi_{(j)} = \frac{n_{(K-j+1)}}{\sum_{k=1}^K n_k}, \quad j = 1, \dots, K \quad (2.6)$$

This ensures that the aggregate penalty contributed by each class is normalized, forcing the hyperplane to pay more attention to minority class samples.

Intra-Class Sample Weighting ($f(x_i)$):

To handle noise within a class, we utilize the Mahalanobis distance [Mohammadi and Sarmad \(2019\)](#). Unlike Euclidean distance, Mahalanobis distance accounts for the covariance structure of the data, making it scale-invariant and more sensitive to statistical deviations. For a sample x_i belonging to class j :

$$d_i = \frac{|\sigma_{park,t}^2 - \mu_j|}{\sigma_j} \quad (2.7)$$

where μ_j and σ_j are the mean and standard deviation of Parkinson variances in class j . We map this distance to a membership value using an exponential decay function:

$$f(x_i) = \exp\left(-\frac{d_i}{\beta}\right) \quad (2.8)$$

Samples that deviate significantly from the class volatility profile (potential outliers) receive a lower membership $f(x_i)$, effectively shrinking their margin and reducing their influence on the decision boundary.

2.3 Theoretical Justification

The selection of the Parkinson estimator within a fuzzy framework offers distinct theoretical advantages. By using an estimator with lower variance than squared returns, the feature space representation becomes more stable [Patton \(2011\)](#). Consequently, the computed Mahalanobis distance d_i is less likely to be distorted by measurement noise. This stability ensures that the fuzzy membership function m_i accurately reflects the true relevance of a sample to its class regime, rather than reacting to random price fluctuations.

Lemma 1 (Adaptive Margin Shaping). *Consider the WFSVM optimization problem with $m_i = \pi_j \exp(-d_i/\beta)$, where d_i is the Mahalanobis distance of the Parkinson feature. The resulting decision boundary minimizes the impact of heteroskedastic noise and corrects for skewed class priors.*

Proof Sketch: The term π_j ensures that the sum of penalties for the minority class is comparable to that of the majority class, correcting the bias in the SVM optimization [Wu et al. \(2009a\)](#). Simultaneously, the exponential decay term reduces the penalty coefficient Cm_i for samples with large d_i (statistical outliers). This effectively creates a "soft" margin that rejects noise while tightly fitting representative samples of each volatility regime.

3 Experimental Setup and Results

To validate the efficacy of the proposed WFSVM framework, we conducted extensive experiments on real-world financial data. The primary objectives were to assess the model's ability to classify volatility regimes and to compare its performance against established baselines.

Data Description:

We utilized daily OHLC data for 20 major companies listed on the S&P 500 index, spanning the period from 2018 to 2023. This dataset includes a diverse range of sectors (Technology, Healthcare, Finance, etc.) to ensure the generalizability of the results. The Parkinson volatility was calculated for each day, serving as the primary feature set.

Baseline Models:

The proposed WFSVM was compared against: (1) Standard SVM (RBF kernel), (2) FPin-SVM (which uses intra-class distance for weighting), and (3) ISFFSVM (which employs adaptive feature weighting). All models were implemented in Python using the 'scikit-learn' library. Hyperparameter optimization was performed using 5-fold cross-validation on the training set, searching over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{0.001, 0.01, 0.1, 1\}$. The dataset was split into training (70%) and testing (30%) sets using stratified sampling to maintain class distribution. To ensure reliability, results were averaged over 10 independent runs with different random seeds.

Results and Discussion:

Table 1 presents the estimated GARCH parameters (mean μ and variance ω) for the analyzed companies, which were used to define the underlying volatility regimes. As observed, companies exhibit significant heterogeneity in their volatility profiles Cifter (2013). For instance, Tesla shows high volatility variance compared to Berkshire Hathaway. This diversity validates the need for a robust classifier capable of handling non-stationary distributions.

The classification and regression performance of the models is summarized in Table 2. The proposed WFSVM framework consistently outperforms the baseline methods across all metrics (MSE, RMSE, MAE, R^2). Notably, WFSVM achieved an R^2 of 0.913, compared to 0.873 for standard SVM. This substantial improvement confirms the hypothesis that the combination of Parkinson volatility features and Mahalanobis-based fuzzy weighting significantly enhances generalization Clark and Baccar (2018).

When comparing WFSVM to other fuzzy variants, the reduction in MSE from 0.059 (ISFFSVM) to 0.053 (WFSVM) highlights the importance of the specific distance metric and class weighting strategy used in our approach Lio and Chen (2007). While FPin-SVM and ISFFSVM improve upon standard SVM,

Table 1: Estimated ARCH parameters (μ, ω) for 20 companies.

Company	μ	ω	Company	μ	ω
Apple Inc.	2.193	27.001	JPMorgan Chase	0.963	8.467
Microsoft	1.515	5.618	Visa Inc.	1.074	5.830
Alphabet	0.993	3.526	Johnson & J.	0.831	2.056
Amazon	1.604	14.759	UnitedHealth	1.476	20.179
Tesla	5.737	23.273	P&G	1.032	44.918
Berkshire	0.513	1.219	Mastercard	0.677	6.519
NVIDIA	3.453	21.909	Disney	1.337	5.223
Home Depot	3.243	22.281	Verizon	1.120	3.935
Netflix	3.627	44.455	Intel	2.033	9.971
Comcast	2.153	13.341	Meta	1.929	6.327

they often rely on geometric distances which can be misleading in the presence of heteroskedasticity. Our use of Mahalanobis distance on Parkinson features provides a statistically more robust measure of sample relevance.

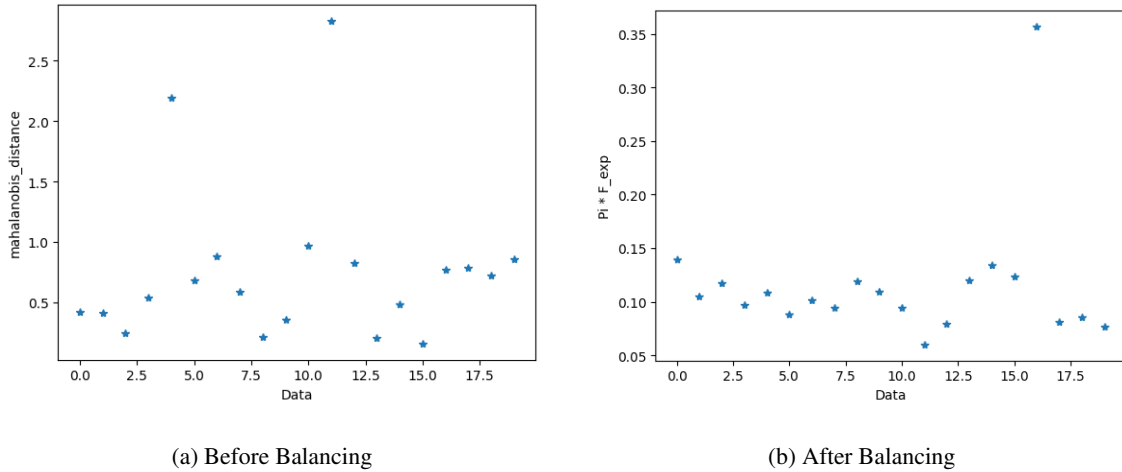


Figure 1: Mahalanobis distance visualization before and after class balancing.

Figure 1 visualizes the effect of the proposed weighting mechanism. The left plot shows the raw distribution of samples, where the majority class dominates the feature space. The right plot illustrates the distribution after applying the inverse frequency weighting π_j and Mahalanobis filtering. The balanced distribution facilitates the formation of a decision boundary that accurately separates minority class samples without being overwhelmed by the majority class noise.

Table 2: Performance comparison (Mean $\pm$ Std, 10 runs).

Model	MSE	RMSE	MAE	R^2
SVM	0.082 ± 0.004	0.286 ± 0.007	0.214 ± 0.006	0.873 ± 0.005
FPin-SVM	0.068 ± 0.003	0.261 ± 0.006	0.196 ± 0.005	0.889 ± 0.004
ISFFSVM	0.059 ± 0.003	0.243 ± 0.005	0.185 ± 0.004	0.902 ± 0.004
WFSVM (Ours)	0.053 ± 0.002	0.230 ± 0.004	0.172 ± 0.003	0.913 ± 0.003

Conclusion

This study proposes a robust WFSVM framework for imbalanced financial time series by integrating Parkinson volatility and Mahalanobis-distance-based fuzzy membership. By effectively addressing both class imbalance and noise, the proposed WFSVM model achieves superior generalization ($R^2 = 0.913$) compared to standard baselines, validating the efficacy of domain-based volatility features for financial classification.

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